THE MASTERMIND COACHING CLASSES

Revision Notes

Class 10 · Science

Chemistry — Chapters 1 to 4 · NCERT Science — Class 10 · CBSE / RBSE Syllabus 2026-27

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Chemical Reactions and Equations	Writing and balancing equations, types of reactions, corrosion and rancidity
2	Acids, Bases and Salts	Indicators, reactions of acids and bases, pH, important salts
3	Metals and Non-metals	Properties, reactivity series, ionic compounds, extraction and corrosion of metals
4	Carbon and its Compounds	Covalent bonding, homologous series, nomenclature, ethanol, ethanoic acid, soaps

The Mastermind Coaching Classes — Road No. 5, VKIA, Jaipur — 9887258679

CHAPTER 01 · CHEMISTRY

Chemical Reactions and Equations

Rusting of an iron gate, souring of milk, burning of a candle and digestion of food are all chemical changes: old substances disappear and new ones form. Chemists describe such changes in a compact language, the chemical equation. This chapter teaches you to write and balance equations, to sort reactions into types, and to see how oxidation harms everyday things through corrosion and rancidity.

Big Questions

- How do we write a reaction so that no atom is lost or created?
- What are the main types of chemical reactions and how do we recognise them?
- Why do iron rust and fatty food turn bad, and how can this be prevented?

1.1 Chemical Change and its Signs

A **chemical reaction** changes the identity of substances. We can usually spot one by observing: a change of **state**, a change of **colour**, evolution of a **gas**, formation of a **precipitate** (an insoluble solid that settles out) or a change in **temperature**.

Observation	Example
Change in colour	Lemon juice added to potassium permanganate solution: purple turns colourless
Gas evolved	Zinc granules with dilute sulphuric acid give bubbles of hydrogen
Precipitate formed	Sodium sulphate + barium chloride gives white barium sulphate
Temperature change	Quicklime in water: the mixture becomes hot
Change of state	Burning candle wax: solid to liquid to gaseous products

1.2 Writing and Balancing Chemical Equations

A **word equation** names reactants and products joined by an arrow. A **skeletal equation** replaces names with formulae. Because atoms can neither be created nor destroyed, the number of atoms of every element must be **equal on both sides**; such an equation is **balanced**.

DEFINITION

Law of conservation of mass: in a chemical reaction the total mass of the reactants equals the total mass of the products. A balanced equation is the symbolic form of this law.

Step 1 — Write the skeletal equation
$Fe + H_2O \rightarrow Fe_3O_4 + H_2$

Step 2 — Count atoms of each element on both sides
Take the compound with the most atoms (Fe_3O_4) first and begin with the element that appears in it.

Step 3 — Balance by writing coefficients
Never alter a subscript (that would change the substance). Put 3 before Fe and 4 before H_2O, then 4 before H_2.

Step 4 — Check and add state symbols
$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$

- State symbols: **(s)** solid, **(l)** liquid, **(g)** gas, **(aq)** aqueous (dissolved in water). A downward arrow marks a precipitate, an upward arrow a gas.
- Conditions (heat, light, catalyst, pressure) are written above or below the arrow; the symbol Δ means heat.
- Most reactions are **not** reversible on paper; a single arrow means reactants are converted to products.

IMPORTANT

Change only the **coefficients** while balancing. Changing a subscript, for instance H_2O to H_2O_2, creates a different compound.

1.3 Types of Chemical Reactions

Type	General form	Example
Combination	A + B → AB	$CaO + H_2O \rightarrow Ca(OH)_2$ + heat; $2Mg + O_2 \rightarrow 2MgO$
Decomposition	AB → A + B	$CaCO_3 \rightarrow CaO + CO_2$ (heat); $2H_2O \rightarrow 2H_2 + O_2$ (electricity)
Displacement	A + BC → AC + B	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
Double displacement	AB + CD → AD + CB	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$
Redox	Oxidation and reduction together	$CuO + H_2 \rightarrow Cu + H_2O$

▸ Combination and exothermic or endothermic changes

Reactions that release heat are **exothermic**: burning of natural gas, respiration (glucose with oxygen) and decomposition of vegetable matter into compost. Reactions that absorb heat are **endothermic**, such as the decomposition of calcium carbonate.

▸ Decomposition by different energy sources

Energy	Reaction	Observation
Heat	$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$	Green crystals turn brown; pungent smell of burning sulphur
Light	$2AgCl \rightarrow 2Ag + Cl_2$ (sunlight)	White silver chloride turns grey; used in black-and-white photography
Electricity	$2H_2O \rightarrow 2H_2 + O_2$	Hydrogen collects at the cathode, double the volume of oxygen

▸ Displacement and the reactivity idea

A more reactive metal pushes a less reactive metal out of its salt solution. Iron nail in blue copper sulphate solution: the solution fades towards green and a brown deposit of copper forms. Zinc displaces copper, and lead displaces copper from copper chloride, but copper cannot displace iron.

▸ Double displacement

The two compounds exchange ions. A **precipitation reaction** produces an insoluble product, for example lead nitrate with potassium iodide gives yellow lead iodide. A **neutralisation** (acid + base) is also a double displacement.

1.4 Oxidation and Reduction

DEFINITION

Oxidation is gain of oxygen or loss of hydrogen. **Reduction** is loss of oxygen or gain of hydrogen. Both occur at once in a **redox reaction**. The substance oxidised is the one that gains oxygen; the substance reduced is the one that loses it.

Reaction	Oxidised	Reduced
$CuO + H_2 \rightarrow Cu + H_2O$	H_2 (gains oxygen, becomes H_2O)	CuO (loses oxygen, becomes Cu)
$ZnO + C \rightarrow Zn + CO$	C	ZnO
$2Cu + O_2 \rightarrow 2CuO$ (black coating)	Cu	O_2

1.5 Effects of Oxidation in Daily Life

▸ Corrosion

When metals are slowly eaten away by air, moisture or chemicals the process is **corrosion**. The reddish-brown coating on iron is **rust** (hydrated iron oxide); the black coating on silver is silver sulphide; the green coating on copper is basic copper carbonate. Rusting needs **both air (oxygen) and water**.

- Prevention: painting, oiling, greasing, galvanising (zinc coating), chrome plating, anodising, making alloys.
- Corrosion damages bridges, ships, railings and car bodies, costing huge sums every year.

▸ Rancidity

Fats and oils in food get oxidised on keeping, giving an unpleasant smell and taste. This is **rancidity**. It is slowed by adding **antioxidants**, packing food in **nitrogen** (as chips are packed), refrigeration and airtight containers.

REMEMBER

Burning and rusting are both oxidation; burning is fast and gives light and heat, rusting is slow.

1.6 Worked Examples

Balance: $Al + O_2 \rightarrow Al_2O_3$. Put 2 before Al_2O_3 (4 Al, 6 O), then 3 before O_2 and 4 before Al: **$4Al + 3O_2 \rightarrow 2Al_2O_3$**.

Identify the type: $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$ is double displacement (a neutralisation); $2KClO_3 \rightarrow 2KCl + 3O_2$ is decomposition.

Why does a copper vessel turn green? Copper reacts slowly with moist air containing carbon dioxide to form a green layer of basic copper carbonate.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Why must a chemical equation be balanced?	To obey the law of conservation of mass: atoms of each element must be equal on both sides.
Why is the burning of natural gas called exothermic?	It releases heat energy to the surroundings.
Name the type: $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$.	Double displacement and precipitation; PbI_2 is a yellow precipitate.
Why does a green solution of iron(II) sulphate crystals turn brown on heating?	It decomposes to iron(III) oxide, which is brown, releasing SO_2 and SO_3.
Why is hydrogen collected in double volume in electrolysis of water?	Water is H_2O: two hydrogen atoms for each oxygen atom.
How is chips packaging protected from rancidity?	The pack is flushed with nitrogen so that oxygen is absent.
What is galvanisation?	Coating iron with a thin layer of zinc to protect it from rusting.
Write the equation for the reaction of quicklime with water.	$CaO + H_2O \rightarrow Ca(OH)_2$ + heat.

★ KEY TAKEAWAYS

- A balanced equation has **equal atoms of every element on both sides**; only coefficients may be changed.
- Four main types: **combination, decomposition, displacement, double displacement**; redox is oxidation and reduction occurring together.
- **Exothermic** reactions release heat, **endothermic** reactions absorb it.
- A more reactive metal displaces a less reactive metal from its salt solution.
- **Corrosion** (needs air and water) and **rancidity** are oxidation effects; prevent them by coating, alloying, antioxidants and nitrogen packing.

CHAPTER 02 · CHEMISTRY

Acids, Bases and Salts

Lemon juice is sour, soap feels slippery, and curd, vinegar and toothpaste all belong to families of substances that behave in predictable ways. This chapter explains how acids and bases are identified with indicators, how they react with metals, carbonates and each other, how strength is measured on the pH scale, and why salts such as common salt, baking soda, washing soda and plaster of Paris matter in everyday life.

Big Questions

- How can we tell an acid from a base without tasting it?
- What happens when acids and bases react, and what is the pH scale?
- Which salts are made from common salt and why are they useful?

2.1 Indicators

An **indicator** shows by a change of colour whether a substance is acidic or basic. **Natural indicators** include litmus, turmeric, red cabbage and petals of China rose; **olfactory indicators** (onion, vanilla, clove oil) change their smell in acid or base.

Indicator	In acid	In base
Blue litmus	Turns red	Stays blue
Red litmus	Stays red	Turns blue
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow
Turmeric	Yellow	Red-brown

DEFINITION

Acid: produces **H^+ (hydrogen) ions** in water. **Base:** produces **OH^- (hydroxide) ions** in water. A base that dissolves in water is an **alkali** (NaOH, KOH, $Ca(OH)_2$).

2.2 Reactions of Acids and Bases

Reaction	Equation	Test or note
Acid + metal	$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$	Hydrogen burns with a pop sound
Base + metal	$2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$	Hydrogen again; only some metals react
Acid + metal carbonate / hydrogencarbonate	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$	CO_2 turns lime water milky

Reaction	Equation	Test or note
Lime water + CO_2	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$	Excess CO_2 makes milkiness disappear: $Ca(HCO_3)_2$ forms
Acid + base (neutralisation)	$NaOH + HCl \rightarrow NaCl + H_2O$	Salt and water only
Acid + metal oxide	$CuO + 2HCl \rightarrow CuCl_2 + H_2O$	Metal oxides are basic
Base + non-metal oxide	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$	Non-metal oxides are acidic

IMPORTANT

Glucose and alcohol contain hydrogen but are **not** acids: they do not release H^+ ions in water. Acid solutions conduct electricity because of free ions; the glow of a bulb depends on the number of ions.

▸ Acids and bases in water

- Dry HCl gas does not change dry blue litmus; ions form only in **water**. $HCl \rightarrow H^+ + Cl^-$. In water H^+ exists as hydronium ion, $\mathbf{H_3O^+}$.
- **Dilution** of acid or base is exothermic: always add acid to water slowly with stirring; adding water to concentrated acid can splash.
- Dilution reduces ions per unit volume, so the solution becomes less concentrated.

2.3 Strength: the pH Scale

DEFINITION

pH is a number from 0 to 14 expressing the concentration of hydrogen ions: **pH < 7 acidic, pH = 7 neutral, pH > 7 basic**. A change of one pH unit changes H^+ concentration tenfold. Universal indicator gives a colour for each pH.

Substance	Approx. pH	Nature
Gastric juice	1.2	Strongly acidic
Lemon juice	2.2	Acidic
Vinegar	3	Acidic
Pure water, common salt solution	7	Neutral
Blood	7.4	Weakly basic
Baking soda solution	8.2	Weakly basic
Milk of magnesia	10	Basic
Sodium hydroxide	14	Strongly basic

▸ pH in daily life

- **Digestion:** the stomach makes hydrochloric acid; excess causes indigestion, relieved by antacids such as magnesium hydroxide.

- **Tooth decay** starts when mouth pH falls below 5.5 as bacteria produce acid from sugar; basic toothpaste neutralises the acid.
- **Soil:** plants need a suitable pH; farmers treat acidic soil with lime.
- **Self-defence:** bee sting (methanoic acid) and nettle sting are soothed by a mild base such as baking soda.

2.4 Salts: Family and pH

A **salt** is formed when an acid reacts with a base. Salts of strong acid and strong base (NaCl) are **neutral (pH 7)**; salts of strong acid and weak base (NH_4Cl) are **acidic**; salts of strong base and weak acid (sodium carbonate) are **basic**. Salts with the same positive or negative ion belong to a **family** (NaCl, KCl; NaCl, Na_2SO_4).

2.5 Important Salts from Common Salt

Sodium chloride is obtained from sea water or rock salt. On passing electricity through its **brine** (chlor-alkali process), three products form: **chlorine** at anode, **hydrogen** at cathode, **sodium hydroxide** near the cathode. $2NaCl + 2H_2O \rightarrow 2NaOH + Cl_2 + H_2$.

Compound	Preparation / formula	Uses
Bleaching powder	$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$	Bleaching cotton, linen, wood pulp; disinfecting drinking water; oxidising agent
Baking soda	$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$	Antacid; baking powder; soda-acid fire extinguisher
Washing soda	$Na_2CO_3 \cdot 10H_2O$ (recrystallising sodium carbonate)	Glass, soap, paper; removing permanent hardness of water
Plaster of Paris	$CaSO_4 \cdot \frac{1}{2}H_2O$ (gypsum heated at 373 K)	Plaster for fractures, toys, decoration, smoothing surfaces

FORMULA

Baking soda on heating: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$. **Plaster of Paris sets:** $CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O \rightarrow CaSO_4 \cdot 2H_2O$ (gypsum). **Gypsum to POP:** $CaSO_4 \cdot 2H_2O \rightarrow CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O$.

- **Water of crystallisation** is the fixed number of water molecules in one formula unit of a salt. Copper sulphate crystals $CuSO_4 \cdot 5H_2O$ are blue; on heating they lose water and become white anhydrous copper sulphate; adding water restores the blue colour.
- Plaster of Paris must be stored in a moisture-proof container because it sets on contact with water.

2.6 Worked Examples

Why does an acid not show acidic behaviour without water? Ions are produced only when the acid dissolves in water.

Baking powder is baking soda with a mild edible acid such as tartaric acid. On heating or mixing with water, CO_2 is released and makes cakes rise and soft.

A solution has pH 4: it is acidic, so blue litmus turns red, and its H^+ concentration is 1000 times that of a neutral solution of pH 7.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What gas is evolved when zinc reacts with dilute sulphuric acid and how is it tested?	Hydrogen; a burning candle brought near it gives a pop sound.
Why does HCl in water conduct but in benzene not?	It gives ions only in water.
What is the pH range of acidic and basic solutions?	Acidic below 7; basic above 7; neutral exactly 7.
Why should a doctor use plaster of Paris for fractures?	It sets into a hard gypsum mass that holds the bone in place.
Write the chemical name and formula of bleaching powder.	Calcium oxychloride, $CaOCl_2$.
How does an antacid work?	It neutralises excess HCl in the stomach.
Why are tooth decay and pH linked?	Acids made by bacteria lower mouth pH below 5.5 and corrode enamel.
What happens when CO_2 is passed in excess through lime water?	First milky $CaCO_3$; then it clears as soluble $Ca(HCO_3)_2$ forms.

★ KEY TAKEAWAYS

- Acids give **H^+**, bases give **OH^-** in water; indicators change colour in them.
- Acid + metal gives **salt + hydrogen**; acid + carbonate gives **salt + water + CO_2**; acid + base gives **salt + water**.
- **pH scale 0 to 14**: below 7 acid, 7 neutral, above 7 base; each unit is a tenfold change.
- Chlor-alkali process gives **NaOH, Cl_2 and H_2**; bleaching powder, baking soda, washing soda and plaster of Paris are made from them.
- Hydrated crystals hold **water of crystallisation**; $CuSO_4 \cdot 5H_2O$ is blue, anhydrous is white.

CHAPTER 03 · CHEMISTRY

Metals and Non-metals

From the iron of a bridge to the copper of a wire and the carbon of a pencil, materials around us are metals or non-metals, and the two groups behave very differently. This chapter compares their properties, explains how metals react with oxygen, water, acids and salts, places them in the reactivity series, shows how ionic compounds form, and describes how metals are extracted from ores and protected from corrosion.

Big Questions

- How do metals differ from non-metals in physical and chemical behaviour?
- How does the reactivity series decide how a metal is extracted?
- Why are ionic compounds different from covalent ones?

3.1 Physical Properties

Property	Metals	Non-metals
Lustre	Shiny surface (metallic lustre)	Dull; iodine is an exception
Hardness	Hard (sodium, potassium are soft and cut with a knife)	Usually soft; diamond is the hardest natural substance
Malleable / ductile	Beaten into sheets, drawn into wires (gold, silver best)	Brittle; break on hammering
Conductivity	Good conductors of heat and electricity; silver and copper best; lead and mercury poor for heat	Poor conductors; graphite conducts
Melting point	High; gallium and caesium melt on a palm; mercury is liquid	Low; bromine is liquid
Sound	Sonorous	Not sonorous
State	Solids except mercury	Solids, liquids, gases

3.2 Chemical Properties of Metals

Reaction	Equation	Remark
With oxygen	$2Cu + O_2 \rightarrow 2CuO$ (black); $4Al + 3O_2 \rightarrow 2Al_2O_3$	Metal oxides are generally **basic**
Amphoteric oxides	$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$; $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$	Oxides of aluminium and zinc react with both acids and bases
With water	$2Na + 2H_2O \rightarrow 2NaOH + H_2$; $3Fe + 4H_2O$ (steam) $\rightarrow Fe_3O_4 + 4H_2$	Na, K react violently; Ca less so; Mg with hot water; Al, Zn, Fe with steam
With dilute acid	$Mg + 2HCl \rightarrow MgCl_2 + H_2$	Hydrogen is rarely evolved with nitric acid, a strong oxidiser; copper does not react with dilute HCl

Reaction	Equation	Remark
With salt solution	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$	A more reactive metal displaces a less reactive one

- Sodium and potassium react so vigorously with oxygen that they are stored under **kerosene**. Aluminium, zinc and magnesium form a protective oxide layer.
- **Anodising** thickens the protective oxide layer on aluminium.
- **Aqua regia** (3 parts conc. HCl : 1 part conc. HNO_3) dissolves gold and platinum.

3.3 The Reactivity Series

DEFINITION

The **reactivity series** arranges metals in decreasing order of reactivity: **K, Na, Ca, Mg, Al, Zn, Fe, Pb, [H], Cu, Hg, Ag, Au.** Hydrogen is included for reference.

Group	Metals	Behaviour
Highly reactive	K, Na, Ca, Mg, Al	Vigorous reactions; extracted by electrolysis
Moderately reactive	Zn, Fe, Pb	React with acids and steam; extracted by reduction
Low reactivity	Cu, Hg, Ag, Au	Do not displace hydrogen; occur free or as sulphides

3.4 Reaction of Non-metals and Ionic Compounds

Non-metals do not give hydrogen with dilute acids. Their oxides are **acidic or neutral** (CO_2, SO_2 acidic; CO, H_2O neutral).

A metal atom (few outer electrons) **loses** electrons to become a positive ion (cation); a non-metal atom **gains** electrons to become a negative ion (anion), reaching the stable octet. The electrostatic attraction between the ions is the **ionic bond**: sodium atom (2,8,1) loses one electron, chlorine atom (2,8,7) gains it; **Na^+ and Cl^-** form NaCl. Magnesium chloride is Mg^{2+} with two Cl^-.

Property of ionic compounds	Reason
Hard, brittle solids	Strong forces hold the ions in a crystal lattice
High melting and boiling points	Much energy is needed to break the strong attraction
Soluble in water, insoluble in kerosene or petrol	Water molecules separate the ions
Conduct only when molten or dissolved	Ions become free to move; solids have fixed ions

3.5 Occurrence and Extraction of Metals

A natural **mineral** that contains enough metal to be profitably extracted is an **ore**. The unwanted earthy impurities are **gangue**. Extraction depends on the position in the reactivity series.

Step 1 — Enrichment of the ore
Remove the gangue, e.g. by hydraulic washing or magnetic separation.

Step 2 — Extract the metal
Low reactivity: **heating alone** (cinnabar HgS gives mercury; $2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$, then $2HgO \rightarrow 2Hg + O_2$). Middle: convert sulphide ore to oxide by **roasting**, or carbonate by **calcination**, then **reduce** with carbon. Top of the series: **electrolysis of molten chloride**.

Step 3 — Refine the metal
Impure metal is purified by **electrolytic refining**: impure metal is anode, pure metal is cathode, salt solution is electrolyte; pure metal deposits on the cathode while impurities settle as anode mud.

Metal	Method	Equation
Zinc	Roast then reduce	$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; $ZnO + C \rightarrow Zn + CO$
Iron (thermit)	Aluminothermy; used to weld rail tracks	$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ + heat
Sodium	Electrolysis of molten NaCl	Cathode: $Na^+ + e^- \rightarrow Na$; anode: $2Cl^- \rightarrow Cl_2 + 2e^-$
Aluminium	Electrolysis of molten alumina	Cathode: $Al^{3+} + 3e^- \rightarrow Al$

FORMULA

Roasting: heating sulphide ore in excess air. **Calcination:** heating carbonate ore in limited air: $ZnCO_3 \rightarrow ZnO + CO_2$. Both then use carbon reduction: **MO + C → M + CO**.

3.6 Corrosion and its Prevention

Iron rusts in moist air; silver tarnishes black; copper turns green. **Prevent** by painting, oiling, **galvanisation** (zinc coating, which protects even if scratched), electroplating and making **alloys**.

Alloys are homogeneous mixtures of a metal with other metals or non-metals. Examples: **brass** (Cu + Zn), **bronze** (Cu + Sn), **solder** (Pb + Sn, low melting point, used for welding wires), **stainless steel** (Fe + Ni + Cr). Pure iron is soft; adding a little carbon makes it hard and strong. Alloys usually conduct less than their pure metals.

3.7 Worked Examples

Why is aluminium used for cooking utensils? It is a good conductor of heat, light, ductile, malleable and protected by its oxide film.

Why does a copper vessel not react with dilute HCl? Copper is below hydrogen in the reactivity series and cannot displace it.

Compare calcination and roasting. Roasting uses excess air on sulphide ores and produces oxide with SO_2; calcination heats carbonate ores in limited air producing oxide with CO_2.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Why are sodium and potassium kept in kerosene?	They react vigorously with oxygen and moisture in air and may catch fire.
Name a metal that is liquid and a non-metal that conducts.	Mercury; graphite.

Likely question	Model answer
What is an amphoteric oxide? Give an example.	An oxide that reacts with both acids and bases, e.g. Al_2O_3, ZnO.
Why do ionic compounds conduct in molten state?	Ions become free to move and carry current.
How is sodium extracted?	Electrolysis of molten sodium chloride; sodium forms at the cathode.
What is galvanisation?	Coating iron or steel with zinc to protect against rust.
State two uses of the thermit reaction.	Welding broken railway tracks and machine parts.
Why is reduction by carbon not used for aluminium?	Aluminium is more reactive than carbon, so carbon cannot reduce its oxide; electrolysis is used.

★ KEY TAKEAWAYS

- Metals are lustrous, malleable, ductile, sonorous, good conductors; non-metals are mostly the opposite.
- Reactivity series: **K, Na, Ca, Mg, Al, Zn, Fe, Pb, H, Cu, Hg, Ag, Au**.
- Metal oxides are **basic** (some **amphoteric**); non-metal oxides are acidic or neutral.
- **Ionic compounds** have high melting points and conduct when molten or dissolved.
- Extraction: low reactivity by **heat**, middle by **carbon reduction**, top by **electrolysis of the molten salt**; refine by electrolysis.
- Prevent corrosion by **galvanising, painting, alloying**.

CHAPTER 04 · CHEMISTRY

Carbon and its Compounds

Food, fuel, fabrics, medicines and living bodies are all built from carbon compounds, although carbon forms only a small part of the Earth's crust. Carbon's special ability to form long chains, rings and strong bonds explains the huge variety. This chapter covers covalent bonding, the types of carbon chains, naming of organic compounds, the chemistry of ethanol and ethanoic acid, and the cleansing action of soaps and detergents.

Big Questions

- Why does carbon form so many compounds?
- How are organic compounds named and classified?
- How do soaps clean and what is special about ethanol and ethanoic acid?

4.1 Bonding in Carbon

Carbon (atomic number 6, electronic configuration 2,4) has four valence electrons. Losing four electrons needs too much energy, and gaining four is difficult for a small nucleus, so carbon **shares** electrons. Sharing a pair of electrons forms a **covalent bond**. In methane CH_4, carbon shares one pair with each of four hydrogens; in O_2 two pairs are shared (double bond); N_2 shares three pairs (triple bond).

Property of covalent compounds	Reason
Low melting and boiling points	Weak forces between molecules
Poor conductors of electricity	No ions are formed
Mostly insoluble in water, soluble in organic solvents	Molecules are non-polar

▸ Why is carbon special?

- **Catenation:** the ability of carbon to join with itself, forming long chains, branched chains and rings.
- **Tetravalency:** four bonds give links with hydrogen, oxygen, nitrogen, sulphur, halogens.
- Strong C–C bonds make compounds stable.

Allotrope	Structure	Property
Diamond	Each C bonded to four C in a rigid 3-D lattice	Hardest natural substance, non-conductor
Graphite	Layers of hexagons, each C bonded to three others	Soft, slippery, conducts electricity
Fullerene (C-60)	Football-shaped spheres of carbon	Discovered by Smalley, Kroto and Curl

4.2 Saturated and Unsaturated Carbon Compounds

Type	Bonds	Examples	General formula
Alkanes (saturated)	All single bonds	CH_4, C_2H_6, C_3H_8	C_nH_{2n+2}
Alkenes (unsaturated)	One double bond	C_2H_4 (ethene), C_3H_6	C_nH_{2n}
Alkynes (unsaturated)	One triple bond	C_2H_2 (ethyne)	C_nH_{2n-2}

Structural isomers: compounds with the same molecular formula but different structures. Butane C_4H_{10} has two: straight-chain butane and branched 2-methylpropane. Rings such as cyclohexane C_6H_{12} and benzene C_6H_6 also exist.

4.3 Functional Groups, Homologous Series and Nomenclature

DEFINITION

A **functional group** is an atom or group of atoms that decides the chemical properties of a compound. A **homologous series** is a family of compounds with the same functional group, the same general formula and similar chemical properties; consecutive members differ by **$-CH_2-$** (14 u).

Class	Functional group	Suffix / prefix	Example
Alcohol	–OH	-ol	Ethanol C_2H_5OH
Aldehyde	–CHO	-al	Ethanal CH_3CHO
Ketone	>C=O	-one	Propanone CH_3COCH_3
Carboxylic acid	–COOH	-oic acid	Ethanoic acid CH_3COOH
Halo-alkane	–Cl, –Br	chloro-, bromo-	Chloropropane

Step 1 — Count carbon atoms
Use the root: meth (1), eth (2), prop (3), but (4), pent (5), hex (6).

Step 2 — Identify the group
Replace the final -e of the alkane with the suffix.

Step 3 — Show unsaturation
'-ene' for a double bond, '-yne' for a triple bond; a prefix names the halogen.

4.4 Chemical Properties

Reaction	Equation	Remark
Combustion	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat + light	Saturated hydrocarbons burn with a clean blue flame; unsaturated give a yellow sooty flame because of incomplete combustion
Oxidation	$CH_3CH_2OH \rightarrow CH_3COOH$	Alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ add oxygen

Reaction	Equation	Remark
Addition (unsaturated)	$CH_2=CH_2 + H_2 \rightarrow CH_3CH_3$ (Ni catalyst)	Used to make vegetable ghee from oils
Substitution (saturated)	$CH_4 + Cl_2 \rightarrow CH_3Cl +$ HCl (sunlight)	One atom replaces another

4.5 Ethanol and Ethanoic Acid

Feature	Ethanol (C_2H_5OH)	Ethanoic acid (CH_3COOH)
Physical	Colourless liquid, soluble in water; neutral to litmus	Colourless liquid; 5 to 8 per cent solution is vinegar; freezes at 290 K (glacial acetic acid)
With sodium	$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$	Also evolves H_2
Special reaction	Dehydration with hot conc. H_2SO_4: $C_2H_5OH \rightarrow C_2H_4 + H_2O$	With carbonates: $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$
Uses	Solvent, medicines, tonics, fuel mixed with petrol	Vinegar, preservative, solvent

FORMULA

Esterification: $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ (conc. H_2SO_4); esters have a sweet fruity smell and are used in perfumes. **Saponification:** ester + NaOH → alcohol + sodium salt of the acid (soap).

- **Ethanoic acid is a weak acid:** it is only partly ionised, unlike HCl which is fully ionised.
- **Drinking alcohol:** methanol is poisonous and can cause blindness; denatured alcohol has methanol and a dye added to make it unfit to drink; alcohol slows the nervous system and harms the liver.

4.6 Soaps and Detergents

A **soap** is a sodium or potassium salt of a long-chain carboxylic acid. Each molecule has a water-loving (**hydrophilic**) ionic head and a oil-loving (**hydrophobic**) hydrocarbon tail. In water the tails point inwards towards the oil and the heads face the water, forming **micelles** that trap grease and are rinsed away; the colloidal solution scatters light.

Soap	Detergent
Salts of fatty acids	Sodium salts of sulphonic acids or ammonium salts
Forms scum (insoluble Ca and Mg salts) in hard water	Works well in hard water; no scum
Biodegradable	Some detergents are non-biodegradable and pollute

4.7 Worked Examples

Name: $CH_3CH_2CH_2CHO$ has four carbons and an aldehyde group: **butanal**. **Formula of hexane:** n = 6, C_6H_{14}. **Number of covalent bonds in ethane C_2H_6:** 7 (six C–H, one C–C).

Distinguish ethanol from ethanoic acid: add sodium hydrogencarbonate; the acid evolves CO_2 with brisk effervescence, ethanol does not.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Why does carbon form covalent bonds?	It has four valence electrons and cannot easily lose or gain four electrons, so it shares them.
What is catenation?	The self-linking property of carbon to form chains and rings.
Why is a sooty flame seen on burning unsaturated hydrocarbons?	Carbon-rich compounds burn incompletely, leaving carbon particles.
Give the molecular formula of the alkane with 5 carbons.	C_5H_{12} (pentane).
Why are detergents better than soaps in hard water?	They do not form insoluble scum with Ca and Mg ions.
Write the product of hydrogenation of ethene.	Ethane, with nickel catalyst.
What is denatured alcohol?	Ethanol made poisonous by adding methanol so it cannot be drunk.
How does a soap clean?	Its tails dissolve in grease and heads in water, forming micelles which are washed away.

★ KEY TAKEAWAYS

- Carbon forms **covalent bonds**; **catenation and tetravalency** give millions of compounds.
- Alkane C_nH_{2n+2}, alkene C_nH_{2n}, alkyne C_nH_{2n-2}.
- A **homologous series** differs by CH_2; the **functional group** fixes the properties.
- Ethanol + ethanoic acid → **ester** (sweet smell); ester + NaOH → soap (saponification).
- **Micelles** explain cleansing by soap; detergents work in hard water.